Pointlike scattering from fermionic color strings: mechanism tests for quark–diquark and lepton–meson correspondences

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Abstract

We study the proposal that bosonic color strings with quark–antiquark or quark–quark endpoints describe mesons or diquarks, while fermionic color strings with quark–antiquark or antiquark–antiquark endpoints describe leptons or quarks. The primary objective is a mechanism making the whole fermionic string appear pointlike in experiments while bosonic strings remain extended. As a first test, a warped effective model with ultraviolet-localized fermionic modes and extended bosonic modes produces a hierarchy of elastic-current form factors. A second test identifies the necessary localization: marginal and infrared-weighted profiles cannot achieve the same separation with a comparable scale hierarchy, and the efficient profiles require ultraviolet boundary or completion physics. The mode profiles are assumed, not derived from a common string construction; one elastic current is not a complete string scattering amplitude. Earlier flat-string and geometric analyses remain comparisons, while contact binding and scalar matching are supporting appendices. A consistent string realization, its endpoint dictionary, and its full experimental scattering criterion remain open.

Provenance and revision. Version 0.1 was reconstructed from detailed conversations before receipt of the earlier preprints and is preserved at Git commit `333e537`. Those preprints have since been received and compared. Version 0.2 makes the four string sectors and pointlike-scattering mechanism central; treating a contact trimer as the main proposal was a drift corrected by the user. Source-informed reconstruction is not blind discovery. Standard warped-field, few-body and SQCD tools are used with explicit assumptions; their use does not itself establish novelty or a completed string theory.

1 The question and its historical target

The proposed correspondence is the following. "Bosonic" and "fermionic" label the string sectors; replacing a string by a third pointlike constituent is a separate model, not an identification assumed here.

State	$SU(3)$ color-string sector	Physical endpoints	Target net color
Meson	bosonic	$q, \bar{q}$	$\mathbf{1}$
Diquark	bosonic	q, q	$\bar{\mathbf{3}}$
Lepton	fermionic	$q, \bar{q}$	$\mathbf{1}$
Quark	fermionic	$\bar{q}, \bar{q}$	$\mathbf{3}$

The diquark entries select the antisymmetric color channels. Internal color flux does not imply that the entire meson or lepton is colored. This table specifies the research target, not a constructed spectrum or a completed charge dictionary. The intended endpoint quarks are vectorlike Dirac fields; weak generations are deferred. Global chiral symmetry remains relevant to the eventual pion sector.

The *main goal* is to explain approximately pointlike scattering of the whole fermionic strings, in contrast to extended bosonic strings. Pointlike endpoint constituents do not answer that question. Section 2 directly tests a current-localization mechanism and its limits. Binding and scalar matching are retained only as supporting studies in Appendices A–B.

The user identifies Schwarz's 1971 *Dual quark-gluon model of hadrons* [1] as a historical anchor, with related early work involving Joel Scherk. Ramond's fermionic dual construction [2], dual parton models, and spinful string endpoints [3] are nearby leads. Before comparing claims, three uses of "dual" must be distinguished: channel duality of resonance amplitudes, a gauge/string description of a confining theory, and an electric–magnetic equivalence between gauge theories. An analogy between these ideas is not a derivation of their equivalence. Primary-text review must identify what the 1971 construction actually assigns to its fermionic and bosonic states.

There is also an immediate representation issue. On an ordinary oriented open-string stack the massless Neveu–Schwarz (NS) vector and Ramond (R) spinor carry the same Chan–Paton representation, ordinarily an adjoint. Calling them "gluon" and "quark" does not turn the latter into a fundamental **3** of color. Projections, multiple stacks, or additional matter sectors can alter this statement, but their consistency and interactions must be supplied. A meson is itself color singlet, whereas a gluon is adjoint. Thus the quark–gluon and lepton–meson identifications are distinct targets even when inspired by the same string spectrum.

We retain two operational levels for comparison. A *weak correspondence* permits an elementary fermion with hadronic dressing. A *strong correspondence* requires the state and its observables to follow from one string or dual construction. The old singlet-meson/carrier, mixing and contact-trimer examples are diagnostics; none substitutes for the four-sector proposal.

For a lepton-like state F the intended scattering criterion is a comparison with a specified point-particle theory over a finite kinematic domain,

$$\left| \frac{d\sigma_F/d\Omega}{d\sigma_{\rm point}/d\Omega} - 1 \right| < \epsilon, \qquad \Lambda_{\rm had} \ll E \le E_*, \tag{1}$$

away from angular singularities or zeros in the reference amplitude. Mass, charge, radiative corrections, and infrared resolution must be matched. For quarks, confinement prevents a literal isolated asymptotic-quark measurement: the eventual criterion must use factorized hard amplitudes and infrared-safe jet or hadron observables. A representation count alone cannot meet Eq. (1).

2 Two tests of a pointlike-current mechanism

2.1 A specified observable and effective model

The ingredients are standard warped bulk-field and holographic-current tools [8, 9]. The present test applies them to a specified pair of sector densities; it does not claim a new localization principle. We first test an ingredient of Eq. (1): the normalized spacelike elastic current of the *whole effective string mode*. Assuming minimal coupling and no independent Pauli or transition form factor, factoring the ordinary spin kinematics from a process with a single such current insertion gives Born cross-section ratio $|F(Q)|^2$ relative to its matched point-particle reference. We use $|F(Q)|^2 \ge 0.99$ as an illustrative tolerance, not an experimental bound. Two dressed vertices, inelastic channels, string resonances and radiative corrections require additional calculations.

Take an AdS slice with $z_0 = \Lambda_{\rm UV}^{-1}$, $z_1 = \Lambda_{\rm had}^{-1}$ and $\rho = z_1/z_0$. A minimally coupled probe with fixed ultraviolet source and Neumann infrared boundary condition has

$$\mathcal{V}(Q,z) = \frac{A(Qz)}{A(Qz_0)}, \qquad A(t) = t[K_1(t) + bI_1(t)], \qquad b = \frac{K_0(Qz_1)}{I_0(Qz_1)}. \tag{2}$$

It obeys $\mathcal{V}(Q, z_0) = 1$ and $\partial_z \mathcal{V}(Q, z_1) = 0$. For $Q > 0$ it is positive and decreases towards the infrared. The corresponding overlap form factor is

$$F(Q) = \int_{z_0}^{z_1} P(z)\mathcal{V}(Q, z)\, dz, \qquad \int_{z_0}^{z_1} P(z)\, dz = 1, \tag{3}$$

so charge normalization gives $F(0) = 1$.

A five-dimensional Dirac field with *nonzero* mass $m_5 = c/\ell_{\mathrm{AdS}}$ can support a massless *four-dimensional* zero mode with canonically rescaled profile $f(z) \propto z^{-c}$. Its norm has weight $dz\,|f|^2$. A vectorlike four-dimensional Dirac state can be formed from two bulk fields with opposite-sign masses and complementary chiral boundary conditions, chosen so that the surviving left and right modes share this profile. For $c > 1/2$, write $\nu = 2c - 1$ and take

$$P_F(z) = \frac{\nu z_0^{\nu} z^{-1-\nu}}{1 - \rho^{-\nu}}, \qquad P_B(z) = \frac{2z}{z_1^2 - z_0^2}. \tag{4}$$

The bosonic density is an extended phenomenological comparison, not a derived QCD meson wavefunction. The fermionic density follows from the stated effective Dirac zero mode; identifying that mode with a Ramond or other physical fermionic string remains to be done. A literally massless five-dimensional Dirac field has $c = 0$ and does not supply the assumed ultraviolet localization.

The proposed mechanism is different sampling of warped energy scales. In a string realization with a common proper string scale, excitations situated at smaller z have a larger four-dimensional energy scale. This motivates the profile test but does not derive the two profiles from one string background. In particular, unbroken supersymmetry constrains actual partner spectra and matrix elements; independently assigning Eq. (4) requires compatible boundary dynamics or supersymmetry breaking.

2.2 Test 1: finite-momentum separation of the currents

Let $x = Qz_0$ and $y = Qz_1$. For fixed $0 < \nu < 2$ and $x \ll 1 \ll y$, integration by parts gives

$$C_\nu = \nu \int_0^\infty t^{-1-\nu}[1 - tK_1(t)]\, dt = \int_0^\infty t^{1-\nu} K_0(t)\, dt = 2^{-\nu}\Gamma(1 - \nu/2)^2. \tag{5}$$

The boundary terms vanish within that range of ν. It follows that

$$1 - F_F(Q) \simeq \frac{C_\nu (Q/\Lambda_{\mathrm{UV}})^\nu - \rho^{-\nu}}{1 - \rho^{-\nu}}, \tag{6}$$

$$F_B(Q) \simeq 4(\Lambda_{\mathrm{had}}/Q)^2. \tag{7}$$

The fermionic expansion has corrections of order $x^2|\log x|/(1 - \rho^{-\nu})$ and exponentially small infrared terms. It is not uniform as $\nu \to 2$ or $\nu \to 0$; those limits require the finite-cutoff expression. These are current overlaps, not the radial average of a flat-string four-point amplitude used later in this paper.

For $c = 1$, $\nu = 1$ and $C_1 = \pi/2$. At $Q = 10\Lambda_{\mathrm{had}}$,

$$1 - F_F \simeq \frac{5\pi - 1}{\rho - 1}. \tag{8}$$

The leading estimate for the one-current tolerance is therefore ρ of order 3×10^3, while $F_B \simeq 0.04$. The finite-momentum calculation below evaluates Eq. (3) rather than treating the asymptotic estimate as an exact threshold.

Numerically integrating the finite-cutoff current gives the examples in Table 1. For $c = 1$, solving $F_F^2 = 0.99$ at $Q = 10\Lambda_{\rm had}$ gives

$$\rho_{\rm threshold} \simeq 2888.50, \qquad F_F = 0.99498744, \qquad F_B = 0.03998718. \tag{9}$$

This is the threshold of the specified model and tolerance, not a lower bound extracted from experiment. At $\rho = 3000$ the fermionic rate ratio is 0.990366; raising the momentum to $100\Lambda_{\rm had}$ at $\rho = 10^4$ instead gives 0.970144. The interval over which the current appears pointlike therefore matters.

ρ	$Q/\Lambda_{\rm had}$	F_F $(c = 1)$	F_F^2	F_B
1000	10	0.985846	0.971892	0.0399953
3000	10	0.995171	0.990366	0.0399871
10000	10	0.998537	0.997076	0.0399859
10000	100	0.984959	0.970144	0.000400094

Table 1: Finite-momentum overlaps for the assumed profiles. The complete scan has 23 points with $Q/\Lambda_{\rm UV} \leq 0.1$. The bosonic column is an amplitude form factor, not its square.

Adaptive quadrature in $\log z$ was compared at the threshold and at $(Q/\Lambda_{\rm had}, \rho) = (100, 10^4)$ with fixed Gauss–Legendre quadrature and independently evaluated Bessel functions. Higher precision Bessel arithmetic and two quadrature orders gave agreement with the adaptive results within 4×10^{-16} in absolute value at those points. The fixed-rule nodes and saved outputs use double precision, so this comparison does not certify 40 or 60 accurate decimal digits. For the next test, the marginal profile $c = 1/2$ at $(Q/\Lambda_{\rm had}, \rho) = (10, 3000)$ gives $F_{1/2} = 0.726914$, well below the $c = 1$ value 0.995171 at the same scales.

There is an exact ordering behind the effect: $P_F/P_B \propto z^{-2-\nu}$ decreases, whereas $\mathcal{V}$ also decreases. The normalized fermionic distribution is more ultraviolet concentrated, giving $F_F \geq F_B$. Identical normalized densities instead give identical form factors in this minimal overlap model. Spin statistics alone do not generate the difference.

2.3 Test 2: the localization and normalizability discriminator

At the marginal value $c = 1/2$, $P_{1/2}(z) = 1/(z \log \rho)$. Its current can be integrated exactly:

$$F_{1/2}(Q) = \frac{K_0(x) - bI_0(x)}{\log \rho\, A(x)}. \tag{10}$$

For $x \ll 1 \ll y$ this becomes

$$1 - F_{1/2} \simeq \frac{\log(Q/\Lambda_{\rm had}) + \gamma_E - \log 2}{\log \rho}. \tag{11}$$

At $Q = 10\Lambda_{\rm had}$, the same one-percent cross-section tolerance requires $\log \rho \gtrsim 436$, or a hierarchy of order 10^{189}–10^{190}. This is an analytic scale estimate, not a numerical fit to data.

For $c < 1/2$ define $\alpha = 1 - 2c > 0$. The normalized density is $P_\alpha = \alpha z^{\alpha-1}/(z_1^\alpha - z_0^\alpha)$. For fixed $\alpha > 0$ and $x \ll 1 \ll y$, keeping the leading cutoff dependence gives

$$F_\alpha(Q) \simeq \frac{2^\alpha \Gamma(1 + \alpha/2)^2 y^{-\alpha} - \rho^{-\alpha}}{1 - \rho^{-\alpha}}. \tag{12}$$

Dropping $\rho^{-\alpha}$ requires $\alpha \log \rho \gg 1$; doing so before taking $\alpha \to 0$ would miss Eq. (11). More strongly, $P_\alpha/P_{1/2} \propto z^\alpha$ increases, so at fixed Q, ρ the exact overlaps obey

$$F_{c<1/2}(Q) \leq F_{c=1/2}(Q). \tag{13}$$

No subcritical member of this family improves on the marginal tolerance requirement.

Finally the unnormalized mode has

$$\|f\|^2 \propto \int_{z_0}^{z_1} z^{-2c}\, dz. \tag{14}$$

This diverges when $z_0 \to 0$ for every $c \geq 1/2$. The efficient profiles thus require a physical ultraviolet boundary or completion; they are not ordinary bulk-normalizable modes after removing that cutoff. Holographically, ultraviolet localization often represents substantial elementary-source admixture. The norm test does not force that interpretation uniquely, but it prevents presenting the mechanism as an established extended composite string made pointlike merely by its fermionic statistics.

2.4 What the two tests establish

The calculation supplies a conditional mechanism for a hierarchy of whole-mode elastic currents. It also identifies the ingredient a stronger construction must explain: the ultraviolet localization and its boundary degrees of freedom. A candidate common string or Seiberg-dual theory must derive those ingredients, implement the four endpoint assignments and their charges, and control other probes, transition amplitudes and string excitations. The present extended bosonic profile is not yet a standard meson's spectrum or pion dynamics. These are explicit remaining construction tasks, rather than consequences silently assumed from the form-factor calculation.

3 Operator parity and the two-branch dictionary

3.1 Physical endpoints and labels are different constructions

With physical Dirac quarks, a gauge-invariant meson interpolator is

$$M^i{}_j(x,y) = \bar{q}_j(y)\, W(y,x)\, \Gamma q^i(x), \tag{15}$$

where W is a bosonic color Wilson line. This operator is even under fermion parity. Gluons and additional $q\bar{q}$ pairs cannot change that parity. For ordinary color-singlet QCD operators at $N_c = 3$,

$$(-1)^F = (-1)^{N_q + N_{\bar{q}}} = (-1)^{N_q - N_{\bar{q}}} = (-1)^{3B} = (-1)^B, \tag{16}$$

where the last equality uses integer baryon number for the color singlet. It does not apply to an arbitrary new sector with independently assigned charges.

An exact duality preserves the physical parity of a state. It may express an odd state as a nonlocal disorder operator or topological excitation, but it cannot identify the even operator in Eq. (15) with an odd elementary field without extra structure. Conversely, Chan–Paton flavor labels are not physical four-dimensional constituent spinors. A model using labels can assign an R state to an endpoint sector, but must then derive, rather than assume, its relation to the QCD operator.

3.2 Color, flavor, and the missing charge carriers

The relevant color products are

$$\mathbf{3} \otimes \overline{\mathbf{3}} = \mathbf{1} \oplus \mathbf{8}, \qquad \overline{\mathbf{3}} \otimes \overline{\mathbf{3}} = \mathbf{3}_A \oplus \overline{\mathbf{6}}_S. \tag{17}$$

Let D_i^a denote a local scalar antidiquark, color antisymmetric and flavor antisymmetric. For three light flavors it transforms as a flavor triplet. In an S wave its spin wavefunction is antisymmetric. Since color

and spin together contribute two minus signs under exchange, flavor must also be antisymmetric to satisfy the Pauli principle. For N_f flavors the scalar channel therefore has $N_f(N_f-1)/2$ flavor components. The symmetric count $N_f(N_f+1)/2$ belongs to a different spin/color assignment, not automatically to the same scalar antidiquark.

The following factorized operators are charge-bookkeeping diagnostics, not the proposed colored fermionic strings. A fermion-odd matrix-valued line operator between endpoints need not factor into a color-singlet meson and a spectator. Suppressing spatial dressing and Lorentz projectors, the diagnostic interpolators have the schematic form

$$\mathcal{O}_Q = D\,\chi_Q, \qquad \mathcal{O}_\ell = M\,\chi_L. \tag{18}$$

Here χ_Q and χ_L are *physical* odd carriers or symbols for specified odd topological sectors; they are not supplied merely by calling a state Ramond. For this factorized diagnostic, a minimal ledger assuming conserved conventional B and L is

Object	color	$(-1)^F$	B	L
$M = q\bar{q}$ (singlet)	**1**	$+1$	0	0
$D = \bar{q}\bar{q}$ (scalar)	**3**	$+1$	$-2/3$	0
χ_Q	**1**	-1	1	0
χ_L	**1**	-1	0	1
$D\chi_Q$	**3**	-1	$1/3$	0
$M\chi_L$	**1**	-1	0	1

Both carriers can be electrically neutral. This makes, for example, a charge -1 meson component times χ_L a charged-lepton interpolator, and a neutral meson component a neutral-lepton interpolator. It neither identifies a generation nor predicts a mass.

For u, d, s the charge matching in the quark branch reads

$$u \leftrightarrow [\bar{d}\bar{s}]\chi_Q, \quad d \leftrightarrow [\bar{s}\bar{u}]\chi_Q, \quad s \leftrightarrow [\bar{u}\bar{d}]\chi_Q. \tag{19}$$

Their antidiquark charges are $2/3, -1/3, -1/3$. If individual vector flavor numbers are also conserved, the missing flavor-number vector is $(N_u, N_d, N_s) = (1, 1, 1)$ in all three equations. Thus the carrier needs $B = 1$, electric charge zero, and strangeness -1. This is a statement about weights; it does not uniquely determine an entire $SU(3)_f$ representation or justify selecting one baryon multiplet.

A single neutral odd carrier with a fixed pair (B, L) cannot fill both rows of Eq. (18). One must introduce different carrier species or sectors, abandon separate conservation, or change what the endpoint labels mean. If only $B - L$ is retained, using an *antilepton* rather than a lepton can align the required $B - L$ shift with the quark branch. That weaker accounting still needs an explicit charge dictionary and a theory of all allowed interactions. It does not preserve separate B and L automatically.

Introducing elementary χ_Q, χ_L is a consistent way to write trial operators in a vectorlike extension, but is not a derivation of the desired states. A baryonic or junction realization of χ_Q can instead give an ordinary quark dressed with an extra $q\bar{q}$ or diquark–antidiquark component. Whether this is a new quasiparticle or just another interpolator for the same quark is decided by correlation functions and poles. Extra carriers also enlarge the spectrum, whose unwanted stable states and couplings require control.

4 A flat open-string benchmark

4.1 Assumptions and the common scattering factor

The charged string-QED construction discussed by Cullen, Perelstein, and Peskin [4] provides a concrete benchmark. We use its charged Chan–Paton embedding and a massless R sector for which the field-theory amplitude is the desired QED one; a neutral Abelian gaugino would not suffice. Internal-index choices or projections must exclude additional massless exchanges. Under these assumptions the same-sign channel takes the form

$$\mathcal{M}_{\rm str} = \mathcal{M}_{\rm QED} V(t,u), \qquad V(a,b) = \frac{\Gamma(1-\alpha' a)\Gamma(1-\alpha' b)}{\Gamma(1-\alpha'(a+b))}, \qquad M_{\rm s}^2 = \frac{1}{\alpha'}. \tag{20}$$

This is a reproduction and crossing of established amplitudes, not a claim of a new four-Ramond amplitude. The relative minus sign in the two fermion-exchange orderings already belongs to $\mathcal{M}_{\rm QED}$; statistics does not cancel V.

For massless scattering, $s+t+u=0$ and

$$t = -\frac{s}{2}(1-\cos\theta), \qquad u = -\frac{s}{2}(1+\cos\theta). \tag{21}$$

Both arguments of $V(t,u)$ are negative in the physical same-sign region. The low-energy expansion follows from the series of $\log\Gamma$:

$$\log V(a,b) = \sum_{n=2}^{\infty} \frac{\zeta(n)}{n} \alpha'^n \left[a^n + b^n - (a+b)^n\right], \tag{22}$$

$$V(t,u) = 1 - \zeta(2)\alpha'^2 tu + \zeta(3)\alpha'^3 stu + O(\alpha'^4 s^4), \tag{23}$$

$$\frac{d\sigma_{\rm str}}{d\sigma_{\rm QED}} = 1 - \frac{\pi^2}{12}\left(\frac{E}{M_{\rm s}}\right)^4 \sin^2\theta + O\Big((E/M_{\rm s})^6\Big). \tag{24}$$

The first oscillator correction in this embedding is dimension eight. Dimension-six interactions can occur in other open-string matter constructions [5]; their absence here is not a universal protection of fermions. Corresponding massless NS amplitudes have the same ordered string factor in the simple common sector. The ten-dimensional NS massless state is a vector; a pion identification requires additional compactification and chiral dynamics.

4.2 The hadronic-scale test

At 90°, with $z = \alpha' s$, Eq. (20) reduces to

$$V_{90}(z) = \frac{\Gamma(1+z/2)^2}{\Gamma(1+z)} \underset{z\to\infty}{\sim} \sqrt{\frac{\pi z}{2}}\, 2^{-z}. \tag{25}$$

Using a representative hadronic slope $\alpha'_{\rm had} = 0.9\,\mathrm{GeV}^{-2}$ gives $M_{\rm s} = 1.05409\,\mathrm{GeV}$. Direct high-precision evaluation gives

E (GeV)	$\lvert V_{90}(\alpha'_{\rm had} E^2)\rvert^2$
0.1	0.9999338174
0.3	0.9949170528
1	0.6651692156
3	$1.797464657 \times 10^{-4}$
10	$9.276323046 \times 10^{-53}$

Consequently this common-slope benchmark cannot describe an apparently elementary fermion at energies well above the hadronic scale. Solving the exact 90° ratio for a one-percent deviation yields $M_s/E = 2.94795$. This is an illustrative sensitivity conversion, not a limit from an experiment. The same-sign factor has no physical resonance poles in this region; crossed channels can expose the excitation spectrum directly.

Quark scattering has additional color orderings and gluon channels. A universal multiplication of every dijet amplitude by the factor in Eq. (20) is unjustified. The benchmark diagnoses a specific mechanism, not every possible hadronic string dual.

5 What warping can and cannot establish

Gauge/string duality can convert flat-space string softness to hard power behavior [6]. To inspect that possibility without asserting a complete background, consider the positive convolution

$$V_\nu(z,x) = \nu \int_0^1 du\, u^{\nu-1} V(zu,x), \quad \nu > 0, \tag{26}$$

$$V(w,x) = \frac{\Gamma(1+wx)\Gamma(1+w(1-x))}{\Gamma(1+w)}, \qquad x = \frac{1-\cos\theta}{2}. \tag{27}$$

This is a normalized mathematical ansatz. Spinor wavefunctions, bulk interactions, redshift factors, and a boundary-current prescription would be required to derive it as a four-dimensional fermion amplitude.

Expanding at small z and changing variables to $w = zu$ at large z give

$$1 - V_\nu = \zeta(2)x(1-x)\frac{\nu}{\nu+2}z^2 + O(z^3), \tag{28}$$

$$V_\nu(z,x) \sim \nu z^{-\nu}\mathcal{I}_\nu(x), \qquad \mathcal{I}_\nu(x) = \int_0^\infty dw\, w^{\nu-1} V(w,x). \tag{29}$$

The integral converges for $0 < x < 1$ and $\nu > 0$. A beta-function representation supplies a separately checkable form,

$$\mathcal{I}_\nu(x) = \Gamma(\nu+1)x(1-x)\int_0^1 \frac{d\xi}{\xi(1-\xi)}\left[-x\log\xi - (1-x)\log(1-\xi)\right]^{-\nu-1}. \tag{30}$$

For $\nu = 2$, $x = 1/2$, its value is 4.64412305369104. At the preceding 10 GeV benchmark, $|V_2|^2 = 1.314914125 \times 10^{-6}$. The suppression is far weaker than the flat exponential, but the amplitude is still not close to the point-particle one.

There is a useful necessary condition within a more general positive amplitude average,

$$R(s,x) = \frac{\mathcal{A}_F}{\mathcal{A}_{\text{point}}} = \int_0^\infty da\, \rho(a) V(as,x), \qquad \rho \geq 0, \quad \int da\, \rho = 1. \tag{31}$$

For fixed $0 < x < 1$, V decreases from one to zero. If $f_{\text{had}} = \int_{\alpha'_{\text{had}}}^\infty da\, \rho(a)$ and $R^2 \geq 1 - \epsilon$, then

$$f_{\text{had}} \leq \frac{1-\sqrt{1-\epsilon}}{1 - V(\alpha'_{\text{had}} s, x)}. \tag{32}$$

At 10 GeV and 90°, one-percent agreement requires $f_{\text{had}} \lesssim 0.00501256$. This is an amplitude-weight bound under a positive factorization ansatz, not a Fock-space probability bound. General amplitudes include phases, transition currents, and interference; their spectral residues need not have this positivity. The $\nu \to 0^+$ limit of Eq. (27) concentrates weight at zero local slope, but is not a derived physical mode.

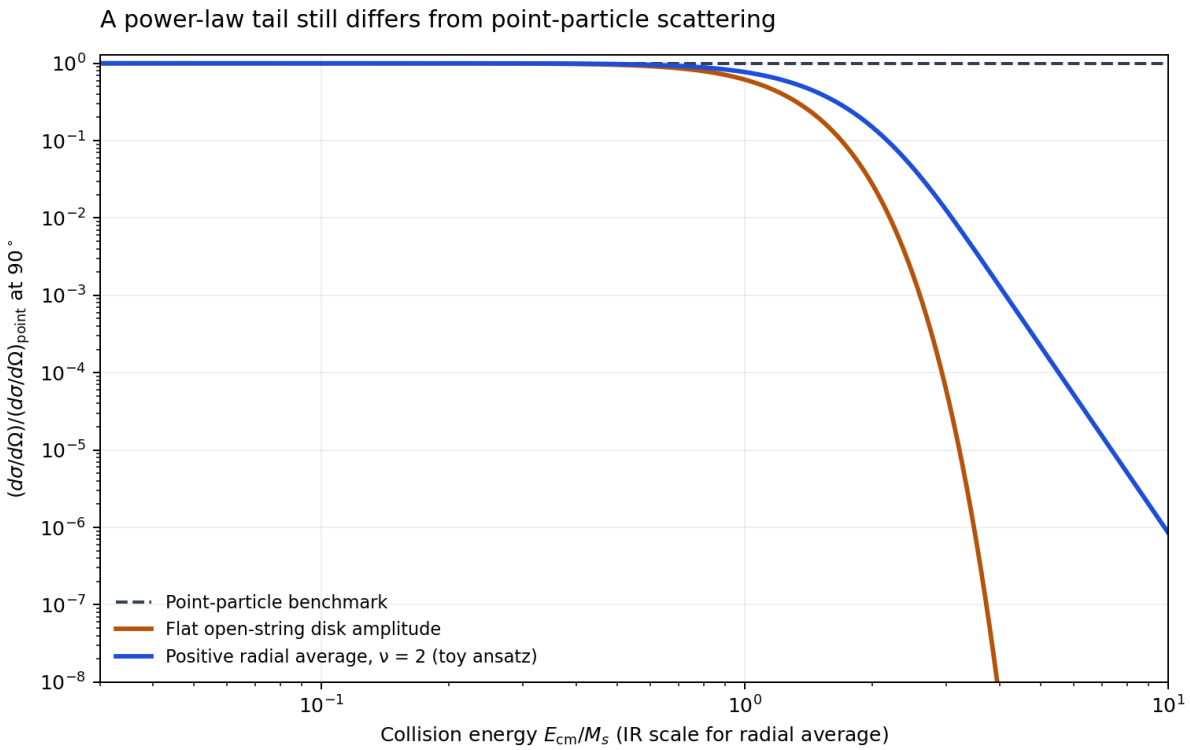


Figure 1: Previously generated notebook comparison of the flat benchmark and the positive radial ansatz at 90°. The reference ratio for a point particle is one. These curves calculate specified functions; they are not a fit or a complete holographic amplitude.

Constituent counting provides a related diagnostic in its perturbative exclusive regime: $d\sigma/dt \sim s^{2-n}$, with n the total number of active minimal constituents. Four elementary external fermions give $n = 4$, whereas four two-constituent states give $n = 8$. An extra resolved odd carrier increases the count again. An anomalous ultraviolet description may change the appropriate operators and counting; it must be calculated. A conformal unitarity bound on a gauge-invariant spinor cannot be applied indiscriminately to a gauge-charged quark field. Inclusive parton scattering and elastic whole-hadron scattering also answer different questions [7].

6 Fermionic coherence, exchange, and contact limits

The motivating intuition contains a valid distinction. If a physical state or density matrix commutes with fermion parity P, then

$$P\chi P^{-1} = -\chi \quad \Longrightarrow \quad \langle\chi\rangle = 0. \tag{33}$$

A fermionic field does not develop an ordinary c-number classical mean field in a parity-preserving state. Formal Grassmann coherent states do not change this conclusion. Even fermion-pair operators can condense, and $\langle T\chi(x)\bar{\chi}(y)\rangle$ remains nonzero. Bosonic binding, moreover, does not require a macroscopic population of on-shell exchanged bosons; in a confining theory a Wilson-loop observable is a better starting point than occupation-number intuition.

An allowed interaction is

$$\mathcal{L}_{\text{int}} = -\sum_i g_i \phi_i^\dagger \bar{\chi}\psi_i + \text{h.c.} \tag{34}$$

It mediates transitions $F_1 + B_2 \leftrightarrow B_1 + F_2$. A source changes its spin when it emits or absorbs the

fermion, so total angular momentum and fermion parity are conserved. A three-fermion cubic vertex for unchanged elastic fermionic sources would instead be odd and absent in an ordinary local parity-even Lagrangian. This selection rule restricts channels, not the range of an allowed propagator.

The free propagator and the zero-frequency spatial Green function are

$$S_F(q) = \frac{i(\not{q} + m)}{q^2 - m^2 + i0}, \qquad G_m(r) = \frac{e^{-mr}}{4\pi r}. \tag{35}$$

The static spinor kernel contains $(m + i\boldsymbol{\gamma} \cdot \boldsymbol{\nabla})G_m$, up to Fourier and overall-phase conventions. In particular,

$$\boldsymbol{\nabla} G_m = -\hat{\boldsymbol{r}} \frac{e^{-mr}}{4\pi} \left(\frac{m}{r} + \frac{1}{r^2} \right). \tag{36}$$

An allowed massless channel can therefore have a power-law spatial kernel; channel projectors can also make a particular matrix element vanish. An off-diagonal kernel $K(r)$ in a two-channel Hamiltonian gives eigenvalues, relative to the mean energy,

$$E_\pm = \pm\sqrt{\Delta^2/4 + |K(r)|^2}. \tag{37}$$

Thus exchange can shift energies without an odd one-point function. A zero-frequency approximation requires negligible local transition frequencies, not merely equality of the two total channel energies. At fixed frequency $|\omega| < m$ the exponential scale is $\sqrt{m^2 - \omega^2}$; above threshold propagating and inelastic channels invalidate a naive static potential.

For $\mathcal{L} = \bar{\chi}(i\not{\partial} - m)\chi - \bar{\chi}J - \bar{J}\chi$, integrating out a heavy χ gives

$$\mathcal{L}_{\text{eff}} = -\bar{J}(i\not{\partial} - m)^{-1}J = \frac{\bar{J}J}{m} + \frac{\bar{J}i\not{\partial}J}{m^2} + \cdots . \tag{38}$$

This expansion is local only for transfers small compared with m; chiral projectors can remove the leading term. An auxiliary fermion without a kinetic term can implement an exact algebraic contact interaction, but is then not a propagating string state. Off-shell propagation does not temporarily violate energy or angular-momentum conservation, and uncertainty relations do not authorize such a violation.

Finally, contact binding does not imply a pointlike bound state. A leading renormalized zero-range nonrelativistic theory with scattering length $a > 0$ has

$$\psi(r) = \frac{e^{-r/a}}{\sqrt{2\pi a}\, r}, \qquad E_B = \frac{1}{2\mu a^2}. \tag{39}$$

For equal constituent masses with one charged constituent, the normalized charge form factor is

$$F(Q) = \frac{4}{aQ} \arctan \frac{aQ}{4} = 1 - \frac{a^2Q^2}{48} + \cdots , \qquad r_{\text{ch}}^2 = \frac{a^2}{8}. \tag{40}$$

When $1/a \ll \Lambda_{\text{UV}}$ the composite is resolvable within the EFT although its force is contact to that accuracy. This counterexample isolates the missing step in the proposed coherence argument.

7 A comparison diagnostic: elementary–composite mixing

A simple effective theory makes the weaker correspondence concrete. Introduce an elementary Dirac fermion F_0 and a composite interpolating resonance C with the same exact unbroken charges. Suppose, as a restricted example, that their Dirac mass matrix is real symmetric,

$$\mathcal{L}_m = -(\bar{F}_{0L}, \bar{C}_L) \begin{pmatrix} m_0 & \delta \\ \delta & M_C \end{pmatrix} \begin{pmatrix} F_{0R} \\ C_R \end{pmatrix} + \text{h.c.}, \qquad M_C > m_0. \tag{41}$$

The light field is $F = \cos\vartheta F_0 - \sin\vartheta C$, with

$$\tan 2\vartheta = \frac{2\delta}{M_C - m_0}, \quad \eta \equiv \sin^2\vartheta = \frac{1}{2}\left[1 - \frac{M_C - m_0}{\sqrt{(M_C - m_0)^2 + 4\delta^2}}\right]. \tag{42}$$

The general Dirac problem has distinct left and right rotations; this symmetric choice is a calculable diagnostic, not a prediction.

For a charged lepton-like state, assume a diagonal probe current in this basis, with no transition form factor, $F_0(Q^2) = 1$, and $F_C(0) = 1$. Then

$$F_F(Q^2) = 1 - \eta + \eta F_C(Q^2), \qquad r_F^2 = \eta r_C^2. \tag{43}$$

The charge at $Q = 0$ is preserved; a small composite overlap suppresses its radius. This positivity and diagonality need not hold for a general current matrix. A single form factor also does not determine all four-point amplitudes, so Eq. (1) still requires a scattering calculation.

For a second diagnostic, let a massive vector resonance couple as

$$\mathcal{L}_\rho \supset \frac{1}{2} m_\rho^2 \rho_\mu \rho^\mu + g_\rho \rho_\mu \bar{C}\gamma^\mu C. \tag{44}$$

The light–light coupling is $g_\rho \eta$. At transfers below m_ρ, eliminating the vector gives

$$\Delta\mathcal{L}_F = -\frac{g_\rho^2 \eta^2}{2m_\rho^2}(\bar{F}\gamma_\mu F)(\bar{F}\gamma^\mu F) + \cdots. \tag{45}$$

There are also heavy–light transition couplings. A small η suppresses an elastic radius by η and this particular four-fermion coefficient by η^2; these are distinct observables. The derivative expansion must not be extrapolated above the resonance, where poles, production, and widths must be included. For illustration only, $g_\rho = 3$, $m_\rho = 1\,\mathrm{GeV}$, and $\eta = 10^{-3}$ give $r_F/r_C = 0.03162$ and $|C_6| = 4.5 \times 10^{-6}\,\mathrm{GeV}^{-2}$. No experimental acceptability is inferred from these selected numbers.

This model exhibits a route to small visible compositeness but inserts the elementary state by hand. For the strong correspondence, a duality must explain why the weakly coupled variable exists, why its hadronic overlap is small, and why its partners have the required spectrum. The small mixing can be technically natural if setting it to zero restores independent symmetries, but this does not establish the desired strong binding. Generic mass mixing can also destabilize a small fermion mass unless its chiral symmetry is specified.

Fermion parity alone cannot provide the suppression. A bosonic operator can couple to $\bar{F}F$ or to $\bar{F}\gamma^\mu F$, and the vector-current square in Eq. (45) is allowed even by a massless fermion's chiral symmetry. Therefore a small Dirac mass, the absence of $\langle\chi\rangle$, and parity selection rules are insufficient to eliminate hadronic resonance couplings. A viable protection must act on the relevant currents, overlaps, or scales and survive radiative corrections.

8 Vectorlike KK ingredients on $\mathbb{CP}^2 \times S^1$

8.1 The spin structure and triplet modes

The geometric avenue is worth separating from the scattering model. Let H generate $H^2(\mathbb{CP}^2, \mathbb{Z})$ with $\int H^2 = 1$. Then

$$c_1 = 3H, \qquad c_2 = 3H^2, \qquad p_1 = 3H^2, \qquad w_2 = H \pmod 2. \tag{46}$$

Thus $\mathbb{CP}^2$ is not spin, and a direct product with S^1 does not remove the obstruction. Its isometry algebra is $\mathfrak{su}(3)$ but its effective scalar isometry group is $PSU(3)$. Ordinary scalar harmonics lie in (p,p), with dimension $(p+1)^3$ and zero triality. They do not supply a fundamental color triplet.

A spin^c structure with determinant line bundle $\mathcal{O}(d)$ requires odd d. Since $\widehat{A} = 1 - H^2/8$, its index is

$$\operatorname{ind} D_d = \int_{\mathbb{CP}^2} e^{dH/2}\widehat{A} = \frac{d^2-1}{8}. \tag{47}$$

The canonical spin^c operator twisted by $\mathcal{O}(k)$ has $d = 2k+3$, so

$$\operatorname{ind} D_k = \frac{(k+1)(k+2)}{2}. \tag{48}$$

For the standard Kähler realization and $k \geq 0$, zero modes are holomorphic sections of $\mathcal{O}(k)$. They transform as $(0,k)$ in a dual-fundamental convention. At $k=1$ there are three components of one color representation, not three generations. Conjugate triplet modes occur in the corresponding negative twist, $k=-4$, through degree-two Dolbeault cohomology. The index theorem by itself does not specify a dynamical gauge embedding.

A nine-dimensional spin^c Dirac field on $M_4 \times \mathbb{CP}^2 \times S^1$, with an internal zero mode and a periodic circle zero mode, can yield a four-dimensional vectorlike Dirac field. No gauge-chiral Standard Model is required for this first step. Nevertheless a nine-dimensional bare mass is allowed unless a further symmetry excludes it; internal zero modes alone do not protect a vanishing four-dimensional mass.

8.2 Charges, KK masses, and the background

At a bookkeeping level choose a quotient generated by $(\omega I, e^{2\pi i/3})$ in $SU(3)\times U(1)$, with $\omega^3 = 1$. A representation of color triality t and integer circle charge n descends when

$$t + n = 0 \pmod 3. \tag{49}$$

With $t=1$, the assignments $n_u = 2$, $n_d = -1$, and $Q = n/3$ give the familiar up/down charges, and their meson differences are $0, \pm 1$. This is a possible charge lattice, not a derived electroweak or compactification embedding.

If these charges are literal momenta around a product circle of radius R_1, the separated Dirac spectrum has

$$m_4^2 = m_9^2 + \lambda_{\mathbb{CP}^2}^2 + \frac{(n+\theta)^2}{R_1^2}, \tag{50}$$

where $\lambda_{\mathbb{CP}^2}$ is a Dirac eigenvalue and θ a common boundary twist. The two assignments obey

$$\min_\theta \max\{|2+\theta|, |-1+\theta|\} = \frac{3}{2}. \tag{51}$$

They cannot both remain light for a high KK scale within this minimal construction. A flux-dependent spectrum, a different gauge combination, extra fields, or different global data would have to overcome this obstruction explicitly. The spin^c connection, a circle graviphoton, and electromagnetism must not be silently identified. In particular, the minimal spin^c charge condition and a physical quark charge ratio of -2 do not trivially fit one common odd-charge assignment.

A nontrivial circle bundle with odd first Chern class can instead make the total space spin; the Hopf case gives S^5. That changes the topology and spectrum. The round positive-curvature S^5 has no ordinary untwisted Dirac zero modes, so the spin obstruction's removal is not enough to solve the light-fermion problem. Finally $\mathbb{CP}^2$ has positive curvature: a flat four-dimensional vacuum times this

space is not a solution of vacuum Einstein equations with zero cosmological constant. Fluxes, stress energy, stabilization, and the relation between gauge couplings, radii, and gravity must be supplied.

The useful outcome is therefore a list of viable geometric ingredients and definite obstructions, not an achieved KK model. A high geometric scale can in principle coexist with a low confining scale through dimensional transmutation. For example, the one-loop six-Dirac-flavor relation

$$\Lambda = M_{\rm KK} \exp\left[-\frac{2\pi}{7\alpha_s(M_{\rm KK})}\right] \tag{52}$$

gives $\alpha_s(M_{\rm KK}) \simeq 0.08296$ for $M_{\rm KK} = 10\,\text{TeV}$ and $\Lambda = 0.2\,\text{GeV}$. This illustrative estimate ignores thresholds and supplies neither a vacuum nor the odd string sector.

9 Chiral dynamics and a dual strong theory

9.1 The pion is a dynamical constraint

For two massless vectorlike flavors, QCD has approximate global $SU(2)_L \times SU(2)_R$ flavor symmetry. A condensate breaks it to the vector subgroup. With $U = \exp(i\tau^a\pi^a/f_\pi)$, the leading chiral action is

$$\mathcal{L}_\pi = \frac{f_\pi^2}{4}\,\text{tr}(\partial_\mu U \partial^\mu U^\dagger) + \frac{f_\pi^2 B_0}{2}\,\text{tr}(\mathsf{m}U^\dagger + U\mathsf{m}^\dagger), \tag{53}$$

where $\mathsf{m} = \text{diag}(m_u, m_d)$. It yields

$$m_\pi^2 = B_0(m_u + m_d) + \cdots \tag{54}$$

before electromagnetic corrections are included. A vectorlike gauge theory can thus provide the required pion mechanism without weak interactions. A common excitation slope, a massless endpoint, or a formally unpaired state of a spectral algebra cannot replace these Ward identities. In particular an unpaired pion in superconformal hadronic spectroscopy [10] is not evidence for a physical negative-orbital-angular-momentum fermion slot.

Current quark masses, dressed constituent masses, effective string endpoint masses, and the whole state's mass are distinct. Massless endpoints do not remove string tension or Regge excitations. The pion radius also contains chiral infrared physics; it should not simply be set equal to $\sqrt{\alpha'}$. A scalar antidiquark and a pseudoscalar meson are different channels, so their assignments must include parity and chiral transformation laws as well as electric charge.

9.2 What a duality would have to deliver

A dual confining description could evade an obstruction tied to the flat benchmark by changing which variables are elementary in the probed regime. A controlled precedent is $\mathcal{N} = 1$ SQCD [11]: an electric $SU(7)$ theory with $N_f = 10$ has a magnetic $SU(3)$ description. The one-loop coefficients are $b_e = 11$ and $b_m = -1$, placing it in the free-magnetic window. This shows that weak magnetic variables can arise from a strongly coupled description; it is not ordinary QCD. Scalars, superpotentials, mesons, and the global-charge dictionary are integral to the example. Removing superpartners or flavors can change the phase and gauge group and must be tracked, not assumed harmless.

There is also a scale-direction issue. A free-magnetic description is useful below its electric strong scale; Eq. (1) demands pointlike behavior above the ordinary hadronic scale. A viable use of this precedent must place the dual matching scale high enough and separately generate the low confining scale, or derive a different ultraviolet completion. Simply labeling an infrared magnetic quark "elementary" does not extend its validity arbitrarily far upward in energy.

A proposed exact correspondence should specify a generating functional with matching external sources, or a controlled approximation to one. At minimum the two descriptions must agree on exact symmetries and their anomalies, charges, fermion parity, physical poles, and current matrix elements in a shared regime. For the operators in Eq. (18), a useful first object is the renormalized matrix correlator

$$G_{ij}(p) = i \int d^4x\, e^{ipx} \langle 0|T\, \mathcal{O}_i(x) \overline{\mathcal{O}}_j(0)|0\rangle, \tag{55}$$

with a gauge-invariant prescription for physical observables or an explicitly fixed gauge for colored interpolators. Shared poles and residues distinguish overlap with one state from an extra particle, but pole matching alone does not prove equality of four-point functions. The candidate must also reproduce the meson spectrum, chiral Ward identities, and the hard-current criterion without extra light sectors contradicting its intended spectrum.

An ordinary baryon is already a fermionic composite bound by bosonic QCD fields. Its existence illustrates why the statistics of a composite does not determine the statistics of its binding channel. Similarly an R string still has the bosonic embedding coordinates and string tension, and four-R scattering can exchange NS states. The desired suppression must therefore be a property of a specified dual dynamics, not a general consequence of the word "fermionic."

10 A falsifiable construction program

The immediate task following Sec. 2 is to derive the different mode profiles and ultraviolet boundary physics from a candidate realization of the four string sectors, then test additional scattering channels. A Seiberg-dual description is one possible source of an emergent weak fermionic variable, but its operator and scale dictionary must be supplied. The contact and scalar-completion studies do not take priority over this pointlike-scattering objective. The broader stages below organize later obligations.

1. **Recover the precise historical target.** Read Schwarz's full 1971 paper and identify the related Scherk papers. Its journal metadata and abstract have been verified, while the supplied preprints have been compared. Preserve the version-0.1 Git checkpoint and separate historical claims from the present construction.

2. **Close the operator dictionary.** Specify whether endpoints are physical partons or labels; choose conserved charges and all carriers; implement Pauli symmetry and chiral representations. A failure to construct an allowed odd operator with the desired charges rejects that assignment before scattering is considered.

3. **Compute a minimal dynamical discriminator.** In a chosen EFT or controlled dual model, calculate a two-point spectral matrix, a probe-current three-point function, and at least one four-point scattering amplitude. Test both the lepton branch and the colored branch with its proper observable. Determine whether the light fermion is inserted, emergent, or merely a dressed known quark.

4. **Explain the scale and residue separation.** Reproduce extended bosonic hadrons while keeping fermion deviations small over a declared interval. If suppression uses mixing or localization, derive it and test loops, transition channels, and resonance production. Contact expansions are tested only in their domain of validity.

5. **Impose the pion mechanism.** Derive the axial Ward identities, the chiral-limit pion, its small explicit mass, and compatible couplings to the odd sector. An unprotected cancellation of unrelated mass parameters does not meet this condition.

6. **Complete a selected ultraviolet route.** For a dual gauge model, follow deformations, anomalies, confinement, and all light fields. For the geometric route, specify the spin^c gauge bundle, solve the vacuum and spectrum, resolve Eq. (50), and compute effective couplings. These are alternative research routes, not established pieces of one common construction.

7. **Match to data and reassess the historical claim.** Derive the full relevant EFT coefficients or finite-energy amplitudes, then apply identified experimental likelihoods with their conventions. Only then compare the model's actual successes with the precise historical proposal. Confirmation cannot be built into the criterion for success.

The current result is a conditional elastic-current mechanism and a discriminator that identifies its dependence on ultraviolet localization. The common string realization and full hard-scattering criterion remain unresolved. The supporting binding example and its failed weak completion neither establish nor exclude that main proposal. Each additional flavor, channel or scale must serve a specified requirement rather than an immediate attempt at all three Standard Model generations.

A A retained adjoint fermion as binding constituent

Consider nonrelativistic fermions $q \in \mathbf{3}$, $\bar{q} \in \bar{\mathbf{3}}$ and $\lambda \in \mathbf{8}$, of masses m, m, M. Color is global and constituent numbers are fixed in this first model. The unique singlet has an octet endpoint pair and

$$T_q \cdot T_{\bar{q}} = \frac{1}{6}, \qquad T_q \cdot T_\lambda = T_{\bar{q}} \cdot T_\lambda = -\frac{3}{2}. \tag{56}$$

The adjoint fermion is an active constituent, not an inert spectator attached to a pre-existing meson. Define a spin-independent short-range density interaction,

$$V = G[T_q \cdot T_\lambda f_R(x) + T_{\bar{q}} \cdot T_\lambda f_R(y)], \quad f_R(r) = \frac{e^{-r^2/R^2}}{\pi^{3/2} R^3}, \quad G > 0, \tag{57}$$

with $x = r_q - r_\lambda$, $y = r_{\bar{q}} - r_\lambda$ and no direct $q\bar{q}$ force. Each density vertex contains an even number of fermion fields. The Hamiltonian assumes a regulated contact coefficient; it does not derive its ultraviolet origin.

With $\mu = mM/(m+M)$, removal of the center of mass gives

$$H_{\text{rel}} = -\frac{\nabla_x^2 + \nabla_y^2}{2\mu} - \frac{\nabla_x \cdot \nabla_y}{M} - \frac{3G}{2}[f_R(x) + f_R(y)]. \tag{58}$$

A Gaussian orbital of width a gives dimer trial energy

$$E_2(a) = \frac{3}{4\mu a^2} - \frac{3G}{2\pi^{3/2}(R^2 + a^2)^{3/2}}. \tag{59}$$

Thus $G > 3\sqrt{3}\pi^{3/2} R/(4\mu)$, obtained at $a^2 = 2R^2$, suffices for a negative exact dimer ground energy ϵ_2. If u is its real normalized orbital, the three-body trial state $u(x)u(y)$ has zero expectation of the recoil cross term, since $\int u \nabla u = 0$. Consequently

$$\boxed{E_3 \leq 2\epsilon_2 < \epsilon_2 < 0.} \tag{60}$$

The relevant breakup threshold is ϵ_2, not zero. Bounded short-range potentials and the variational principle therefore establish a three-body bound state. Removing λ removes both forces. Coupling the

two endpoint spins to zero and retaining the carrier spin gives an odd spin-1/2 state; the species are distinct.

This proof assumes the fixed-number global-color model and the chosen spin-independent interaction. Take $mR, MR \gg 1$ and binding energies small compared with the masses for nonrelativistic control. It supplies neither a flux tube nor stability against newly allowed relativistic channels. Its size is generally nonzero: in the symmetric Gaussian product and a constituent-current approximation, a charged state with neutral λ has

$$\langle r_{\rm ch}^2 \rangle_{\rm trial} = \frac{(m+M)^2 + m^2}{(2m+M)^2} \frac{3a^2}{2}. \tag{61}$$

This is a trial-state radius, not an exact hard-scattering prediction.

B First completion test: scalar exchange and spin recoupling

This supporting test asks whether a specified Yukawa theory generates the interaction needed in Eq. (60). It does not identify the three-body state with the fermionic string. Take one chiral fundamental scalar per endpoint flavor and a Majorana adjoint, with

$$\mathcal{L}_Y = -y\phi^\dagger T^a \psi \lambda^a + \text{h.c.} \tag{62}$$

The endpoint is part of a massive vectorlike Dirac fermion. We use canonically normalized nonrelativistic spinors, for which each massive Weyl particle component is the corresponding spinor divided by $\sqrt{2}$. Additional chiral scalars must be matched separately. With global color this is a Yukawa completion involving a bosonic scalar. In SQCD, $y = \sqrt{2}g$ and gauge-boson interactions also remain; neither interpretation is an exclusively fermionic ultraviolet theory.

B.1 Threshold denominators and the Majorana crossed graph

At zero relative velocity define

$$D_s = m_\phi^2 - (m+M)^2, \qquad D_u = m_\phi^2 - (m-M)^2 = D_s + 4mM. \tag{63}$$

Replacing both by m_ϕ^2 before deciding the mass hierarchy would erase the possible near-threshold enhancement. Let P_0, P_1 project the endpoint–carrier spins onto singlet and triplet, and $P_{\rm swap} = P_1 - P_0$. The connected threshold potentials, before color projection and multiplying a regulated contact, are

$$V_s = -\frac{|y|^2}{2D_s}(T^b T^a)P_0, \qquad V_u = +\frac{|y|^2}{4D_u}(T^a T^b)P_{\rm swap}. \tag{64}$$

The s channel annihilates and recreates the pair in spin zero. The relative crossed sign can also be obtained by normal ordering the number-changing Majorana vertex product: its connected four-field part is minus the spin-swap operator. A conserved Dirac-carrier charge can change which diagrams exist, but requires a fresh charge/channel specification.

In the total singlet the endpoint–carrier pair is a **3** or $\bar{\mathbf{3}}$. The two color tensors have eigenvalues $C_F = 4/3$ and $C_F - C_A/2 = -1/6$, respectively. The resulting pair potential is

$$V_{q\lambda} = -[g_0 P_0 + g_1 P_1] f_R, \tag{65}$$

$$g_0 = \frac{2|y|^2}{3D_s} - \frac{|y|^2}{24D_u}, \qquad g_1 = \frac{|y|^2}{24D_u}. \tag{66}$$

The antiquark endpoint has the same projected coefficients when the masses and couplings agree. For $m_\phi > m + M$, both channels attract, with $g_0 \geq 15 g_1$. Far above threshold,

$$g_0 \simeq \frac{5|y|^2}{8m_\phi^2}, \qquad g_1 \simeq \frac{|y|^2}{24m_\phi^2}. \tag{67}$$

This is not the spin-independent density interaction originally assumed. Here f_R is an EFT regulator; its Gaussian width is not a derived scalar-exchange range. In particular, the s channel is a conversion channel, not a spatial Yukawa potential of range m_ϕ^{-1}.

B.2 A changed three-body spin problem

For three spin-1/2 constituents,

$$P^0_{q\lambda} + P^0_{\bar{q}\lambda} = \frac{1}{2} - \mathbf{S}_{q\bar{q}} \cdot \mathbf{s}_\lambda. \tag{68}$$

Its eigenvalues in $(S_{q\bar{q}}, J) = (0, 1/2), (1, 1/2), (1, 3/2)$ are $1/2, 3/2, 0$. For equal spatial overlaps, the strongest attraction therefore selects endpoint spin one and total spin one-half. The common carrier cannot be in a full singlet with both endpoints simultaneously.

Let $\epsilon_2(c)$ be the ground energy of $p^2/(2\mu) - c f_R$. A product orbital with the strongest fixed spin state gives, when that orbital is bound,

$$E_3 \leq 2\epsilon_2\left(\frac{3g_0 + g_1}{4}\right), \qquad E_{\text{breakup}} = \epsilon_2(g_0). \tag{69}$$

Equation (69) does not automatically lie below breakup whenever a dimer binds. Spatially varying pair weights can mix endpoint-spin sectors, so the fixed-spin product is a variational subspace, not an exact diagonalization. The original proof remains correct for its original Hamiltonian; it cannot simply be transplanted into this completion.

B.3 A sufficient no-binding bound in the weak heavy-scalar regime

Failing a sufficient variational binding test would not prove absence of a state. A separate lower bound supplies that conclusion in a restricted regime. For $h_\nu = p^2/(2\nu) - g_0 f_R$, the central-potential bound [12] gives

$$N_\ell < \frac{2\nu}{2\ell + 1} \int_0^\infty r g_0 f_R(r)\, dr = \frac{\nu g_0}{(2\ell + 1)\pi^{3/2} R}. \tag{70}$$

If the last expression at $\ell = 0$ is below one, h_ν has no negative-energy spectrum. For the three-body recoil operator and $g_0 \geq g_1 \geq 0$,

$$T_{\text{rel}} \geq \frac{p_x^2 + p_y^2}{2m}, \qquad V \geq -g_0[f_R(x) + f_R(y)]. \tag{71}$$

The kinetic inequality follows from $1/\mu - 1/M = 1/m$. Hence

$$\boxed{\frac{m g_0}{\pi^{3/2} R} < 1 \quad \Longrightarrow \quad H_{\text{rel}} \geq h_m(x) + h_m(y) \geq 0.} \tag{72}$$

This excludes Borromean binding as well as dimer-assisted binding in this regulated model. Perturbative y, $m_\phi \gg m, M$, and a nonrelativistic cutoff $\Lambda_c = R^{-1} \ll \min(m, M)$ satisfy Eq. (72) parametrically. The result concerns the matched weak, far-detuned implementation, not arbitrary running contact coefficients or all scalar completions.

B.4 The surviving near-threshold question

For positive detuning $\delta = m_\phi - m - M$,

$$D_s = 2(m+M)\delta + \delta^2, \qquad E \ll \delta \iff p \ll \sqrt{2\mu\delta} \tag{73}$$

is required for an energy-independent contact expansion near threshold. The enhanced coefficient cannot be extrapolated while ignoring this shrinking domain, scalar self-energy dressing, or the $\phi\bar{q}$ and $\bar{\phi}q$ conversion channels. Negative detuning puts such a channel below the free three-fermion threshold. A resonant two-channel theory remains a possible continuation of this supporting study; Eq. (66) does not already establish its binding. The main program instead follows the string-current mechanism in Sec. 2.

Relativistic interactions also change constituent numbers. For equal endpoint flavor, $q\bar{q} \to \lambda\lambda$ can open a three-carrier final sector when masses and charges permit. Choosing distinct exactly conserved endpoint flavors can forbid this annihilation into flavor-neutral carriers. This is a concrete reason to add a second flavor to a completion; it is unrelated to three weak generations. Even after a stable bound pole is obtained, its current and its relation to a super-QCD string remain separate calculations.

B.5 Seiberg variables and the intended odd operator

In vectorlike $\mathcal{N} = 1$ SQCD, the electric meson superfield and its fermion are

$$M_e = \widetilde{Q}Q, \qquad \psi_{M_e} = \widetilde{\phi}\psi + \widetilde{\psi}\phi. \tag{74}$$

The bilinear $\widetilde{\psi}\psi$ instead occurs in its auxiliary/top component. Thus the ordinary mesino is not, by definition, the intended two-fermion endpoints plus gluino. The latter admit

$$\mathcal{H}_\alpha = (\widetilde{\psi}T^a\psi)\lambda^a_\alpha, \qquad R(\mathcal{H}) = 2r - 1 = R(\psi_{M_e}), \quad r = 1 - \frac{N_c}{N_f}. \tag{75}$$

Flavor and baryon number also match. A relation $\mathcal{H} \to c_H\Lambda^3\psi_{M_c} + \cdots$, where $M_e \sim \Lambda M_c$, is symmetry-compatible but uncalculated. The source and renormalization prescription matter; an overlap is not a constituent probability. Holomorphic data alone do not determine the needed scattering matrix elements.

Electric $SU(3)$ with $N_f = 4$ is an s-confining laboratory [11], with

$$W = \frac{\widetilde{B}M_eB - \det M_e}{\Lambda^5}. \tag{76}$$

Its mesons and baryons become free in the infrared; its magnetic gauge group is trivial. At the origin the massless scalar mesons are not already QCD Goldstone pions. With $N_f = 5$, magnetic $SU(2)$ instead lies in the interacting conformal window, with $\Delta(M) = 6/5$ and $\Delta(\psi_M) = 17/10$. Four flavors therefore serve phase control, not a generation count. A useful eventual hierarchy would be $\Lambda_{\text{had}} \ll Q \ll \Lambda$; duality does not establish pointlike scattering above its matching scale, and the lower string/binding scale still needs a deformation.

C Reproducibility and evidence boundaries

All calculations use natural units. The notebook is in the repository directory

`discussions/2026-10-11-quark-antiquark-open-string/`

It contains verbatim copies of the two supplied transcripts with checksums, curated notes, a working blackboard, this manuscript, and reproducible calculations. The scripts `scattering_checks.py`, `geometry_checks.py`, `fermion_exchange_checks.py`, and `branch_checks.py` verify, respectively, the string/radial functions, index and charge algebra, exchange/contact counterexamples, and the new branch/mixing bookkeeping. Results are stored as JSON, with the scattering figure and table alongside them. Checks of equations do not establish a complete vacuum, an exact duality, or agreement with experiment.

The new finite-momentum question in Sec. 2 is implemented in `warped_current_experiment.py`, with its conventions, scan, threshold and numerical comparisons saved in `calculations/warped_current_experiment.json`. Routine implementation and execution were delegated to GPT-6-Sol, following the user's preference for a smaller model for numerical work. The analytic profile discriminator and the supporting binding/completion results were reviewed by additional agents; no numerical physics script was used for those derivations. Agreement between models is not independent physical verification. The notebook records assumptions, corrections and source exposure, with a short milestone table and an idea bank. Its AI-research bibliography distinguishes research reports from proof verification and benchmark performance; FunSearch [13] is a relevant precedent for separately specified evaluators, not a validation of the present theory.

Network access was restored on 12 October. Relevant sections of Seiberg's paper were retrieved, Schwarz's journal metadata and abstract were verified, and Bargmann's publication metadata was checked. The arXiv metadata and abstracts for Refs. [8, 9] were retrieved; the finite-cutoff formulas here were derived directly. Schwarz's full paper and most older pointers below remain unread in this session. No current strongest compositeness limit is asserted. Original uploads and the pre-PDF manuscript checkpoint remain preserved in Git.

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